Speech Science - Lectures 5-6: Speech Science (30 Oct, combined recording)

Session context

the combined deck covers Lecture 05 speech perception on slides 1-23 and begins Lecture 06 speech synthesis at its cover on slide 24. The audio remains one recording; no exact audio cut is inferred.

Lecture overview

This lecture series examines the physical properties governing consonant perception and the historical evolution of speech synthesis, emphasizing how acoustic cues function under noisy conditions. Building on Miller and Nicely's foundational work, the material establishes that listener errors are non-random, revealing that voicing and nasality are the most robust features in noise, while high-pass filtering eliminates these cues and low-pass filtering disrupts place of articulation. The discussion addresses the "invariance problem," noting that coarticulation prevents a one-to-one mapping between acoustic signals and phonemes, yet listeners achieve categorical perception by relying on formant relationships rather than absolute frequencies. Multidimensional Scaling (MDS) is presented as a method to map these perceptual spaces, where frequently confused sounds cluster closely, highlighting the distinction between robust and fragile cues essential for speech recognition diagnostics.

The second portion of the lecture traces the development of speech synthesis from early VODER and formant synthesizers to modern rule-based systems like Klatt's model, all grounded in the source-filter theory. These systems generate speech by combining excitation sources (voiced pulses or noise) with resonators arranged in either cascade or parallel configurations; while cascade structures are simpler for vowels, parallel structures better handle fricatives and plosives despite requiring complex phase management to avoid antiresonances. Formant synthesis is highlighted for its efficiency and lack of storage requirements, relying on dynamic parameter updates every 5 milliseconds to capture rapid speech changes. Key parameters in the Klatt model, such as voicing (AV), frication (AF), and resonator amplitudes (A2F-A6F), allow researchers to precisely manipulate acoustic properties for studying speech identity, though achieving high naturalness historically required extensive manual tuning.

Slide 1: Speech Science 5 2026-2027

This lecture introduces the course "Speech Science 5" for the 2026-2027 academic year. The session is led by Professor Astrid van Wieringen from the Department of Neurosciences at KU Leuven, specifically within the Experimental ORL division. As an introductory slide, it establishes the course identity and instructor credentials without providing further technical content or definitions.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 2: Aim: to examine the perception of consonants under noisy conditions, determine which physical properties of the speech sound affect intelligibility

The study asks which physical properties of speech affect consonant intelligibility under noise; the slide stresses that listener errors are structured rather than random. Results can inform speech perception and recognition, diagnosis, rehabilitation monitoring, and comparisons of speech-processing strategies. Open-set tests collect free responses but are harder to score; closed-set tests offer fixed choices but allow chance guessing.

Slide 3: • Basic test requirements:

Basic test requirements include sensitivity, validity, and reliability. Sensitivity is the ability to detect performance differences, such as varying degrees of hearing loss or psychophysical capabilities, while validity ensures a test measures its intended target and serves as a prerequisite for sensitivity. Reliability refers to measurement repeatability, where random errors inherent in any measurement can be minimized by repeating stimuli within the same test.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 4: 16 consonants /p, t, k, b, d, g, f, ……n /

This slide outlines the analytical tests by Miller and Nicely (1955), which utilized 16 consonants spoken in the /ɑ/ context by five female talkers, with each syllable judged approximately 250 times. The study manipulated speech-to-noise ratios ranging from -18 dB to +12 dB, where positive values indicate a stronger signal than noise and negative values indicate the reverse. Additionally, specific frequency response conditions were tested at a constant +12 dB ratio across various bandwidths to analyze consonant recognition under different acoustic constraints.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 5: Possible setup for closed-set testing

Closed-set testing setups involve presenting stimuli acoustically via loudspeaker or electrically through cochlear implant coils, directly to electrodes, or with a speech processor. In manual protocols where a speaker talks into a microphone and listeners wear headphones, the total number of trials must always equal 250. Unlike these manual methods where data from M & N (1955) may be lost, computerized self-paced testing ensures complete data retention.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 6: • Closed set, 12 repetitions of each stimulus

Slide 6 illustrates a closed-set identification experiment where each of the 16 stimuli is presented 12 times, resulting in a diagonal score of 37% correct identifications against a 6.3% chance level. While the matrix reveals specific error patterns, the slide emphasizes that the distribution of these errors is more informative than the overall accuracy rate alone. The transcript clarifies this methodology by referencing Patricia Kuhl's work on mixed stimuli, where participants identify sounds like /ra/ or /la/, noting that such experiments often display categorical switches in perception rather than gradual…

Evidence: slide content and aligned teacher audio.

Slide 7: Results Miller and Nicely (1955)

This page reproduces Miller and Nicely's consonant-identification confusion matrices for several signal-to-noise conditions. Rows and columns compare presented consonants with listeners' responses; diagonal entries are correct identifications, while off-diagonal counts are confusions. As the signal becomes noisier, response patterns change, showing that errors carry information about which phonetic contrasts listeners can still hear.

Slide 8: Results Miller and Nicely (1955)

This page continues the Miller and Nicely (1955) results with a matrix for a different signal-to-noise and frequency-response condition. Read each cell as a response count, not a probability unless normalized. The tables support the lecture's point that consonant errors are structured by acoustic and phonetic cues, and this block belongs to the combined Lectures 5-6 deck.

Slide 9: • Voicing and nasality is least affected by noise (-12 dB)

Miller and Nicely (1955) demonstrated that listeners detect phonemes using specific cues rather than the whole sound, with voicing and nasality being the most persistent features even in noise (-12 dB). While manner of articulation is generally identified better than place, filtering disrupts these channels differently: high-pass filtering eliminates voicing and nasality causing random confusions, whereas low-pass filtering removes high-frequency frication cues but leaves phonemes audible yet confused. Consequently, disturbing the place of articulation channel prevents distinguishing sounds like /p/, /t/, and /k/.

Evidence: slide content and aligned teacher audio.

Slide 10: 10

Multidimensional scaling (MDS) analyzes complex speech sounds within a perceptual space without prior assumptions about category numbers, revealing that frequently confused sounds cluster closely while rarely confused ones are spaced apart. To ensure sufficient data, MDS is typically performed on summed confusion matrices from multiple subject groups rather than individual matrices with sparse off-diagonal scores. The resulting stimulus space is interpreted through visual inspection and by correlating acoustic properties with the stimuli's coordinates along different dimensions.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 11: 11

Research on speech perception focuses on whether neural mechanisms are species-specific and how the system flexibly handles inherent acoustic variability from factors like coarticulation and speaker differences. Because no one-to-one relationship exists between the acoustic signal and perception, researchers face challenges including the linearity problem, lack of acoustic-phonetic invariance, and segmentation. This flexibility is critical during early development, as delayed auditory exposure beyond age four makes acquiring spoken language significantly more difficult despite potential alternative learning methods like signing.

Evidence: slide content and aligned teacher audio.

Slide 12: 12

Linearity requires each phoneme to map to a specific sound stretch with consistent acoustic properties, but coarticulation violates this by embedding adjacent phoneme information within single waveform segments. Consequently, acoustic-phonetic invariance is absent because identical phonemes vary acoustically depending on context, such as /b/ sounding more rounded before /u/ than /a/, while identical cues can yield different percepts. This variability and lack of invariant mapping make segmenting waveforms into discrete linguistic units extremely difficult.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 13: 13

Listeners perceive constant vowel identity despite acoustic variability across speakers and coarticulation because perception relies on formant relationships rather than absolute frequencies. While the Formant Ratio Theory suggests intervals between F1 and F2 are crucial, it is insufficient alone since some vowels share ratios; thus, Intrinsic models utilize internal spectral patterns, whereas Extrinsic models apply a contextual frame of reference based on speaker characteristics. Additionally, the dynamic movements into a vowel from preceding consonants are equally critical for defining its identity.

Evidence: slide content and aligned teacher audio.

Slide 14: 14

Slide 14 presents research by Strange et al. (1983, 1989) regarding the identification of specific conditions compared to unmodified CVC structures, focusing on how different acoustic cues contribute to vowel identity. The slide includes an illustration of the stimuli used in these studies but provides no further textual details or professor commentary to elaborate on the specific conditions or acoustic mechanisms involved.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 15: 15

Plosives are distinguished by second formant transition loci, ranging from 650-950 Hz for /b/-like sounds and 1500-1800 Hz for /d/-like ones. In an identification experiment, seven synthetic stimuli with equal physical steps are presented in random order, requiring subjects to categorize each as /ba/ or /da/. Results show a sharp shift where the first stimulus is identified as 100% /ba/ and the seventh as 100% /da/, illustrating categorical perception despite the continuous acoustic variation.

Evidence: slide content and aligned teacher audio.

Slide 16: 16

Categorical perception occurs when listeners perceive a continuous acoustic range as discrete categories, showing high discrimination across category boundaries (e.g., /b/ vs. /d/) but chance-level performance within them. The category boundary represents the point of maximum slope where adjacent categories are equiprobable, meaning equal physical differences are not perceived equally. This phenomenon is language-dependent; for instance, Japanese speakers often fail to distinguish /r/ and /l/ because their phonology lacks separate categories for these sounds, whereas American English speakers perceive them distinctly.

Evidence: slide content and aligned teacher audio.

Slide 17: 17

In natural speech, consonantal contrasts rely on multiple acoustic features whose salience varies by context, dialect, rate, and style. Listeners integrate these cues into a unitary percept through phonetic trading relations, where one cue compensates for another; for instance, lower F1 onset frequencies require longer VOT values to identify voiceless stops, while /r/ versus /l/ distinctions depend on the interaction between F3 onset and F1 steady-state duration. This integration allows listeners to categorize sounds despite variability, though perception remains constrained by native phonological categories.

Evidence: slide content and aligned teacher audio.

Slide 18: 18

The McGurk effect demonstrates cross-modal cue integration where conflicting visual and auditory speech stimuli create a perceptual illusion, such as hearing /na/ when seeing /ta/ while hearing /ma/. This phenomenon suggests either a specialized speech system referencing articulatory gestures or general cognitive mechanisms resolving ambiguous multimodal input. As illustrated by the "baba" versus "dada" demonstration, perception shifts based on visual cues, confirming that speech processing is a multimedia event dependent on integrating both modalities rather than auditory information alone.

Evidence: slide content and aligned teacher audio.

Slide 19: 19

Phoneme category boundaries shift based on phonetic context and speaking rate as listeners integrate acoustic information over time. In Mann & Repp's (1980) study, intermediate [s]-[ʃ] stimuli were more likely identified as [ʃ] when followed by [i] than by [u], because listeners compensate for the frequency-lowering effect of [u]'s lip rounding. Similarly, preceding context influences perception; a [da]-[ga] continuum was more often heard as [g] after [əl] than [ər], demonstrating that perceptual boundaries adjust to account for articulatory and acoustic variations in different environments.

Evidence: slide content and aligned teacher audio.

Slide 20: 20

Listeners employ compensatory perceptual processes to decode phonetic messages despite acoustic variations caused by speaking rate changes. In Miller and Liberman's (1979) study, as vowel duration increased to simulate slower speech, the phoneme boundary for a [b-w] continuum shifted toward longer, more gradual transitions. This demonstrates that the brain normalizes perception based on context; faster rates lead listeners to categorize shorter transitions as stops, while slower rates allow longer transitions to be perceived as the same category.

Evidence: slide content and aligned teacher audio.

Slide 21: 21

Listeners interpret phonetic contrasts by analyzing multiple cues within a broader speech context spanning several syllables, rather than relying on isolated acoustic patterns. This process involves sensitivity to coarticulatory and timing influences, suggesting humans possess specialized perceptual mechanisms that recover linguistic messages from underlying articulatory gestures. Furthermore, the brain adapts to a speaker's specific style and rate, demonstrating that perception depends on contextual integration beyond individual sound segments.

Evidence: slide content and aligned teacher audio.

Slide 22: 22

Categorical perception and context effects occur in both speech and nonspeech analogues, as well as in animals, indicating that no distinct neural mechanisms exist specifically for phonetic sounds. Instead, general auditory processing principles handle these signals, while higher-order cognitive factors like expectations and social interaction determine the mode of perception. Although the underlying mechanisms are shared across species, the specific cognitive processing required to categorize sounds remains species-specific, relying on advanced skills found in humans and some animals.

Evidence: slide content and aligned teacher audio.

Slide 23: 23

Linguistic-phonetic categories are likely identified by matching signals against statistical prototypes stored in long-term memory, which are language-specific and adaptable through continuous exposure. While speech perception shares biological roots with general perception, it is a unique, self-contained system that integrates acoustic cues with non-acoustic information like visual mouth movements and prior knowledge. Acoustic input remains primary when clear, but the brain prioritizes the clearest available modality during ambiguity, as demonstrated by the McGurk effect and phonetic restoration where missing sounds are filled in.

Evidence: slide content and aligned teacher audio.

Lecture 06 - speech synthesis (source slides 24-54)

Slide 24: Speech Science 6 2026-2027

This lecture session covers Speech Science 6 for the 2026-2027 academic year. It is presented by Professor Astrid van Wieringen from the Department of Neurosciences at KU Leuven, specifically within the Experimental ORL division. As no specific technical content or definitions were provided in the aligned sources for this slide, it serves solely as an introductory title card identifying the course and instructor.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 25: Faculteit, departement, dienst …

Creating a talking robot requires combining hardware and software while explicitly defining its purpose and scope, unlike human development which evolves naturally. Developers must determine the required level of interactivity, ranging from simple informational responses to full conversations that demand higher processing capabilities. Finally, the system needs speech recognition (Speech-to-Text) with Natural Language Understanding to process input, and Speech Synthesis (Text-to-Speech), potentially using emotional models, to generate expressive, human-like output.

Evidence: slide content and aligned teacher audio.

Slide 26: Faculteit, departement, dienst …

Creating a talking person requires a dialogue management system that handles intent recognition, response generation, and conversation flow through either predefined libraries or flexible AI models. To ensure consistent and engaging interactions, the robot must possess a defined personality, utilize machine learning for adaptive context retention, and maintain memory across multiple exchanges. This process also demands audiovisual integration, with this course specifically focusing on speech acoustics and science to explore the various types of speech generation.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 27: Developments of the past to give context

Building on known human speech production mechanisms, this section explores the historical development of artificial speech synthesis, which expanded significantly after World War II. Early approaches attempted to replicate speech either by modeling articulators or by implementing specific speech cues like prosody and pitch contours. While generating sound became feasible in the 1950s through formal and concatenative methods based on source-filter theory, true discourse generation remains a challenge requiring brain-like capabilities beyond simple signal synthesis.

Evidence: slide content and aligned teacher audio.

Slide 28: Developments of the past to give context

Artificial speech has been a centuries-old dream, with early mechanical efforts dating to 1779 when Christian Kratzenstein in St. Petersburg created acoustic resonators mimicking the human vocal tract. Using vibrating reeds similar to musical instruments and bellows acting as lungs, he successfully generated five long vowels by manipulating physiological differences. While these basic mechanical steps established the scientific foundation for speech production and perception, they predated modern electrical circuits and the computational power required for today's data-intensive synthesis methods.

Evidence: slide content and aligned teacher audio.

Slide 29: Vienna 1791, Wolfgang von Kempelen introduced his "Acoustic-Mechanical Speech Machine", which was able to produce single

In 1791 Vienna, Wolfgang von Kempelen introduced an "Acoustic-Mechanical Speech Machine" capable of producing single sounds and combinations using a pressure chamber, vibrating reed, and deformable leather tube to simulate the vocal tract. By manipulating the tube's shape for vowels and employing constricted passages with a hinged tongue and movable lips for consonants, Kempelen established the theory that the vocal tract is the primary site of acoustic articulation. Later, in the mid-1800s, Charles Wheatstone constructed a more complex version capable of generating full words by utilizing turbulent flow for consonants…

Evidence: slide content; no teacher passage was aligned to this page.

Slide 30: History

Speech synthesis history began with Willis (1838), who used tube resonators to prove vowel quality depends on vocal tract length, not diameter. Stewart's 1922 electrical device introduced the source-filter model using a buzzer and resonant circuits to generate static vowels, though it produced metallic-like sounds lacking intelligibility. Homer Dudley's 1939 VODER advanced this by allowing manual control of voicing and formants via filters, demonstrating artificial speech potential despite requiring significant skill and yielding poor quality. The professor notes that early formant synthesis created the distinct "metallic" voice later used by…

Evidence: slide content and aligned teacher audio.

Slide 31: History

The history of speech synthesis traces back to the VODER and early formant synthesizers like Lawrence's parallel PAT (1953) and Fant's cascade OVE series, which established the source-filter model using vibrating or noise sources. While researchers debated whether to model the vocal tract transfer function in parallel or cascade configurations, John Holmes eventually achieved high naturalness by manually tuning a parallel synthesizer. The professor emphasizes that students need not memorize specific dates but must understand how these systems utilized spectrogram cues and physics to reconstruct speech from individual time-based parameters.

Evidence: slide content and aligned teacher audio.

Slide 32: History

The history of speech synthesis began with George Rosen's 1958 articulatory synthesizer and evolved through mid-1960s Linear Predictive Coding (LPC) experiments, which enabled low-cost but poor-quality systems like the TI Speak'n'Spell. While Noriko Umeda developed the first full English text-to-speech system in 1968, Dennis Klatt's 1987 formant synthesis work established the rule-based foundation for modern systems by defining specific language and phoneme rules. Although early optical reading aids like Kurzweil's were effective, they remained commercially inaccessible due to high costs until increased computational power allowed for more sophisticated signal generation.

Evidence: slide content and aligned teacher audio.

Slide 33: Low level synthesis: Formant synthesis - Klatt synthesis -Parametric synthesis

Formant synthesis, exemplified by Klatt synthesis, generates speech using a source-filter model that simulates sound sources and formant frequencies without modeling physical vocal tract characteristics. Excitation signals, ranging from voiced tones to unvoiced noise or mixed combinations, are filtered through resonators arranged in either cascade or parallel structures. While the simpler cascade structure suits non-nasal voiced sounds, it struggles with fricatives and plosives; conversely, the parallel structure handles these better but requires summing adjacent outputs in opposite phases to prevent unwanted antiresonances.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 34: • The advantage of formant based synthesis is that it does not require a large storage capacity nor high computational demands.

Formant-based synthesis offers the distinct advantage of requiring neither large storage capacity nor high computational demands, making it efficient for research into acoustic properties that determine speech identity. This method relies on a source-filter model where parameters must be updated dynamically, typically every 5 milliseconds, to capture the rapid changes in speech sounds. While historically significant and capable of producing intelligible speech, as demonstrated by Stephen Hawking's synthesizer, modern implementations have evolved to handle complex filter configurations like cascading for improved quality.

Evidence: slide content and aligned teacher audio.

Slide 35: AV= voicing, AF= frication, FL = for flutter (jitter). Set to 20% during the vowel; A2F - A6F are the amplitudes of the parallel

In this model, AV represents voicing, AF denotes frication, and FL stands for flutter (jitter), which is set to 20% during vowel production. The parameters A2F through A6F define the amplitudes of the parallel resonators. Crucially, these specific amplitude values are utilized exclusively when the sound source is identified as frication.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 36: Formant Synthesis

Formant synthesis applies the source-filter theory by generating sound through voiced pulses or noise, which filters then modify to create vowel resonances and consonant characteristics. Systems like DecTalk adjust parameters such as fundamental frequency, formant bandwidths, and breathiness to convey gender and age. This method models only filtering effects, typically requiring five resonant circuits to accurately reproduce male voice formants up to 5000 Hz based on acoustic energy distribution.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 37: How good can formant synthesis be?

Formant synthesis quality depends on patience, as demonstrated by Holmes's natural-sounding results, making it invaluable for research requiring precise parameter control that natural voices cannot provide, such as specific fundamental frequency shifts or allophone transitions. However, its commercial viability is limited by unnaturalness stemming from a lack of subtle variations in parameters and the difficulty of mimicking laryngeal voice sources during rapid F0 changes. This trade-off offers exact experimental values at the cost of requiring extensive manual updates every five milliseconds to identify every allophone.

Evidence: slide content and aligned teacher audio.

Slide 38: Tekst-to-speech (TTS) synthesis: two main phases: high- and low-level synthesis

Text-to-speech (TTS) synthesis comprises two main phases: high-level text analysis, which converts input text into phonetic representations with prosodic details like intonation and stress, and low-level synthesis, which generates the final speech waveforms from this acoustic information. While the course focuses on the low-level generation due to the linguistic complexity of the high-level phase, understanding the full pipeline is essential for recognizing the intermediate steps required to transform typed text into spoken words.

Evidence: slide content and aligned teacher audio.

Slide 39: Different types of synthesis

Pre-recorded speech offers high-quality, non-metallic audio ideal for fixed-vocabulary applications like car systems, weather updates, and assistive devices for individuals with limited communication abilities. However, this method requires significant memory storage and involves a trade-off between quality, capacity, and computation time, leading some to argue it is merely reproduction rather than true synthesis. While unrestricted text-to-speech necessitates shorter units like phonemes or diphones, articulatory synthesis remains impractical due to the extreme complexity of modeling over 200 intricately coordinated muscles.

Evidence: slide content and aligned teacher audio.

Slide 40: Low-level synthesis: Articulatory synthesis

Articulatory synthesis theoretically offers the most accurate artificial speech by directly modeling human articulators and vocal cords as tube sections, often using 2-D X-ray data despite the vocal tract's 3-D nature. While this method accurately captures transients from abrupt area changes better than formant synthesis, its extreme complexity regarding tongue movement rules prevents commercial realization, though tools like Praat demonstrate its feasibility for understanding speech anatomy.

Evidence: slide content and aligned teacher audio.

Slide 41: Problems in Low Level Synthesis

Low-level synthesis methods face distinct challenges: articulatory synthesis demands high computational power to imperfectly model vocal mechanics; formant synthesis struggles with complex rule sets and unnatural nasalized sounds; and concatenative synthesis requires time-consuming data collection while risking distortion at sample boundaries. Additionally, synthesis difficulty varies by language, ranging from the regular pronunciation of Finnish to the irregularities of French, while modern systems also contend with high training energy costs and limitations in adapting recorded voices to different demographics like children.

Evidence: slide content and aligned teacher audio.

Slide 42: Concatenative synthesis

Concatenative synthesis creates natural speech by connecting prerecorded utterances, though it typically requires significant memory and is limited to a single voice. The core challenge involves selecting unit lengths: longer units like demisyllables or diphones preserve coarticulation and naturalness, whereas shorter units save space but risk distortion if transitions are ignored. Since no full TTS system currently relies solely on words or syllables due to complex cutting exceptions like "upkeep," modern systems primarily utilize phonemes, diphones, or demisyllables to balance versatility with acoustic quality.

Evidence: slide content and aligned teacher audio.

Slide 43: Concatenative synthesis

Concatenative synthesis constructs speech by joining recorded units, with diphones often preferred over phonemes because they capture transitions between adjacent sounds. Defined as extending from the steady-state center of one phone to the next, diphones allow concatenation in stable signal regions, reducing distortion and eliminating the need for explicit coarticulation rules. While this approach yields natural consonant clusters and segment durations, it can still struggle with certain syllable boundaries, such as the /pk/ transition in "upkeep," though it generally outperforms demisyllables in these cases.

Evidence: slide content and aligned teacher audio.

Slide 44: Statistical parametric speech synthesis

Statistical parametric speech synthesis (SPSS) generates speech from scratch by statistically modeling waveform parameters rather than storing actual audio or simulating physical articulators like the tongue. Unlike unit selection methods, which require significant effort and offer limited voices, SPSS treats speech as a factorial signal where only the statistical model is stored. This approach enables personalized voice synthesis at a modest cost using just a few sentences of data to create a custom model.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 45: Faculteit, departement, dienst …

Slide 45 introduces the organizational hierarchy terms "Faculteit" (Faculty), "departement" (Department), and "dienst" (Service/Division) within the university structure. The slide displays these concepts alongside the number 22, but no accompanying transcript or detailed explanation is available to define their specific roles, relationships, or operational distinctions in this context.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 46: Faculteit, departement, dienst …

Statistical Parametric Speech Synthesis (SPSS) systems comprise three modules: text analysis, which normalizes input and converts it to linguistic features requiring accurate labels; parameter prediction, which estimates acoustic features like F0 and duration using either HMMs with decision trees or more efficient DNNs that leverage long-term context; and speech synthesis, which generates the final waveform. While HMMs map features to probability densities, DNN-based methods directly map linguistic inputs to acoustic outputs, improving quality by modeling frame correlations, though sequence-to-sequence networks like Wave-Tacotron can bypass intermediate steps to generate waveforms directly…

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Slide 47: Faculteit, departement, dienst …

Statistical Parametric Speech Synthesis (SPSS) operates through training and synthesis phases where a statistical model, rather than stored audio, predicts acoustic features like F0 to generate speech via a vocoder. While Deep Learning-based SPSS offers superior naturalness by utilizing complete context information and distributed representations, it demands extensive corpora and computing resources, risks overfitting with insufficient data, and remains highly dependent on accurate phoneme labeling. As the professor emphasizes, this approach represents a shift toward understanding complex models like Hidden Markov Models to achieve high-quality synthesis, though challenges persist particularly…

Evidence: slide content and aligned teacher audio.

Slide 48: Faculteit, departement, dienst …

Deep learning TTS systems rely on statistical prediction models that are computationally intensive and require extensive, high-quality tagged datasets, making them impractical for smaller developers. While these models offer superior commercial quality compared to traditional methods, they still struggle with out-of-vocabulary words, unnatural prosody, and capturing nuanced emotions without significant fine-tuning. Furthermore, the ability to generate highly realistic speech raises critical ethical and security concerns regarding misuse for deepfakes and impersonation, highlighting that this powerful technology remains imperfect and resource-heavy.

Evidence: slide content and aligned teacher audio.

Slide 49: The first task faced by any TTS system is the conversion of input text into linguistic representation, usually called text-to-phonetic

The initial task for any Text-to-Speech (TTS) system is converting input text into a linguistic representation, known as text-to-phonetic or grapheme-to-phoneme conversion. This process involves identifying the language and parsing the text before addressing pronunciation irregularities caused by historical evolution or borrowed words, such as the variable voicing of "th" in English. To handle these complexities, systems utilize a combination of letter-to-sound rules and lexicons containing morphs for exceptional cases, ultimately generating linguistic data that guides prosodic features like intonation, stress, and duration.

Evidence: slide content and aligned teacher audio.

Slide 50: The parser (syntactic analysis) must

The parser performs syntactic analysis by dividing text into pause-free phrases and determining word emphasis, relying on a grammar and structural mechanism to assign a single, unambiguous interpretation even when input is ambiguous. While full clause-level parsing of unrestricted text remains unsolved, phrase-level parsing is fast and reliable, resolving ambiguities that otherwise hinder the assignment of fundamental frequency and duration in systems lacking semantic context.

Evidence: slide content and aligned teacher audio.

Slide 51: Flowchart of operation of DECtalk, based on rules by Dennis Klatt (formant

DECtalk operates using Dennis Klatt's formant synthesis rules and features seven built-in voices: three male, three female, and one child. The system prioritizes searching its internal dictionary for words before applying letter-to-sound rules to any unrecognized entries. If the resulting pronunciation is unsatisfactory, users can manually input phonemic symbols instead of standard spelling, ensuring the system remains highly intelligible.

Evidence: slide content; no teacher passage was aligned to this page.

Slide 52: Performance evaluation

Performance evaluation assesses speech intelligibility, defined as the listener's ability to distinguish individual sounds, which is critical for applications aiding visually impaired individuals or those unable to speak, such as Stephen Hawking. Measurement involves comparing dictation results against predetermined texts, though listeners often use linguistic knowledge to fill gaps in grammatical sentences. To isolate true recognition capabilities, studies contrast meaningful sentences with syntactically correct but semantically unpredictable ones, noting that listeners increasingly rely on semantic constraints as speech intelligibility decreases.

Evidence: slide content and aligned teacher audio.

Slide 53: Performance evaluation

Performance evaluation distinguishes intelligibility (decoding individual sounds) from comprehensibility (understanding overall content), which is tested via accuracy, response time, and certainty ratings on unfamiliar topics to prevent knowledge bias. While intelligibility focuses on understanding despite poor quality, comprehensibility requires sustained attention that pleasantness or naturalness supports; these subjective criteria are rated on a nine-point scale but remain challenging due to factors like dialects and articulation effects. Ultimately, speech can be understood yet unpleasant, or pleasant yet machine-like, highlighting the complexity of achieving both clarity and natural sound.

Evidence: slide content and aligned teacher audio.

Slide 54: • Speech synthesizers should have the following features:

Ideal speech synthesizers must deliver highly intelligible, natural-sounding voices that adapt to noisy conditions and support multilingual, multidialectal communication. Crucially, they require advanced prosody control and emotional expression to reflect meaning, as emotions drive human decision-making and motivation. While current systems achieve basic understanding, significant challenges remain in achieving true flexibility, naturalness, and audio-visual coherence, particularly regarding complex dialects and context-aware personality traits.

Evidence: slide content and aligned teacher audio.